1. Deriving Stein's estimator

For a constant $q > 0$ we now consider an estimator of the form

$$\hat{\mu} = \Big(1 - \frac{\sigma^2 q}{\|y\|^2}\Big) y. \tag{1.1}$$

- We obtain this estimator with $q = n$ by following the previous idea of minimizing the empirical risk over $s \in \mathbb{R}$.
- This estimator with $q = n - 2$ is called the James-Stein estimator. We will see next that computing the MSE of the James-Stein estimator shows that the MLE is inadmissible for $n \geq 3$.

1.1. **Gaussian integration by parts.**

In order to study the MSE of estimator (1.1), we must first derive several identities related to the normal distribution.

Lemma 1.1. *If $Z \sim N(0, \sigma^2)$ and $f : \mathbb{R} \to \mathbb{R}$ is differentiable such that $\mathbb{E}[|f(Z)|] < +\infty$ and $\mathbb{E}[|f'(Z)|] < +\infty$. Then*

$$\mathbb{E}[Zf(Z)] = \sigma^2 \mathbb{E}[f'(Z)].$$

Proof. Assume $\sigma^2 = 1$ first. This is just an integration by parts: of $\phi(u) = e^{-u^2/2}/\sqrt{2\pi}$ is the standard normal probability density function then

$$\int_{-\infty}^{+\infty} f(z) z \phi(z) dz = \Big[-\phi(z) f(z)\Big]_{-\infty}^{+\infty} + \int_{-\infty} f'(z)\phi(z) dz.$$

The bracket is zero if f has sub-exponential growth at $\pm\infty$ which proves the claim for $\sigma^2 = 1$ in this case. Using only the integrability assumptions on f, f', we can proceed as follows instead: write $f(z) = \int_0^z f'(t)dt$. Then by Fubini's theorem with $I\{Z > 0\}$ the indicator function of event $\{Z > 0\}$,

$$\begin{aligned}
\mathbb{E}[Zf(Z)I\{Z > 0\}] &= \int_0^\infty z\phi(z)\Big(\int_0^z f'(t)dt\Big)dz \\
&= \int_0^\infty f'(t)\Big(\int_t^\infty z\phi(z)dz\Big)dt \\
&= \int_0^\infty f'(t)\phi(t)dt \\
&= \mathbb{E}[I\{Z > 0\}f'(Z)]
\end{aligned}$$

and the same argument for $I\{Z \leq 0\}$ completes the proof.

For $\sigma^2 \neq 1$, with $G = Z/\sigma$ the desired identity can be rewritten as $\mathbb{E}[G\tilde{f}(G)] = \mathbb{E}[\tilde{f}'(G)]$ for $\tilde{f}(u) = f(\sigma u)$. □

Lemma 1.2. *If $y \sim N(\mu, \sigma^2 I_n)$ and $g : \mathbb{R}^n \to \mathbb{R}$ is differentiable with $\mathbb{E}[|(\partial/\partial y_i)g(y)|] < +\infty$ then*

$$\mathbb{E}[(y_i - \mu_i)g(y)] = \sigma^2 \mathbb{E}\Big[\frac{\partial g}{\partial y_i}(y)\Big].$$

Proof. Let us fix one single coordinate $i \in \{1, ..., n\}$ and argue conditionally on the remaining $(n - 1)$ independent variables $(y_l)_{l \neq i}$. For simplicity, assume that $i = 1$, the argument being easily adapted if $i > 1$. Let $Z = (y_1 - \mu_1)/\sigma$ and $f(z) = g(\mu_1 + \sigma z, y_2, ..., y_n)$. An application of the previous lemma gives conditionally on $y_2, ..., y_n$,

$$\mathbb{E}\big[(y_1 - \mu_1)g(y)\big|y_2, ..., y_n\big] = \int \phi(z)zf(z)dz = \int \phi(z)f'(z)dz = \mathbb{E}\big[(\partial/\partial y_1)g(y)\big||y_2, ..., y_n\big]$$

since $f'(z) = (\partial/\partial y_1)g(\mu_1 + \sigma z, y_2, y_3, ..., y_n)$. The proof is complete by the tower property of conditional expectation, i.e., $\mathbb{E}[\mathbb{E}[U|W]] = \mathbb{E}[U]$ for two random vectors U, W (here, $W = (y_2, ..., y_n)$). □

Lemma 1.3. *If $y \sim N(\mu, \sigma^2 I_n)$ and $f : \mathbb{R}^n \to \mathbb{R}^n$ is differentiable with $\mathbb{E}[|(\partial/\partial y_i)f_i(y)|] < +\infty$ for each $i = 1, ..., n$ then*

$$\mathbb{E}[(y-\mu)^T f(y)] = \sum_{i=1}^n \mathbb{E}[(y_i - \mu_i)f_i(y)] = \sigma^2 \mathbb{E}\Big[\sum_{i=1}^n \frac{\partial f_i}{\partial y_i}(y)\Big].$$

Proof. This is simply n applications of the previous lemma. □

1.2. **Stein's Unbiased Risk Estimate.** In this section, it will be useful to write an estimator $\hat{\mu}$, which is a function of the observed vector $y \in \mathbb{R}^n$, as a deterministic function $f : \mathbb{R}^n \to \mathbb{R}^n$ applied to y:

$$\hat{\mu} = f(y).$$

For linear estimators of the form $\hat{\mu} = Ay$ for a deterministic matrix A, the previous sections provided the empirical risk estimate

$$\hat{r}(A) = \|\hat{\mu} - y\|^2 + 2\sigma^2 \operatorname{trace}[A] - \sigma^2 n.$$

The goal of this section is to extend this estimate $\hat{\mu} = f(y)$ for nonlinear functions $f : \mathbb{R}^n \to \mathbb{R}^n$. Note that if $f(y) = Ay$ then the divergence $\sum_{i=1}^n (\partial/\partial y_i) f_i$ is the trace of A which appears in the second term.

For nonlinear f, we have the following generalization which is known as *Stein's unbiased risk estimate.*

Theorem 1.1. *For an estimator $\hat{\mu} = f(y)$ for some function $f : \mathbb{R}^n \to \mathbb{R}^n$ satisfying the assumptions of Lemma 1.3,*

$$\hat{r}_f = \|y - f(y)\|^2 + 2\sigma^2 \sum_{i=1}^n \frac{\partial f_i}{\partial y_i}(y) - n\sigma^2$$

satisfies $\mathbb{E}_\mu[\hat{r}_f] = \mathbb{E}_\mu[\|\mu - \hat{\mu}\|^2]$.

Proof. By developing the square, $\|\hat{m}u - y\|^2 = \|\hat{\nu} - \mu\|^2 - 2\varepsilon^T(\hat{\mu} - \mu) + \|\varepsilon\|^2$ so that

$$\hat{r}_f = \|\hat{\mu} - \mu\|^2 + 2\Big[-\varepsilon^T(f(y) - \mu) + \sigma^2 \sum_{i=1}^n \frac{\partial f_i}{\partial y_i}(y)\Big] + \Big[\|\varepsilon\|^2 - n\sigma^2\Big].$$

The first term is equal in expectation to the MSE of $\hat{\mu} = f(y)$. The second term (first bracket) has zero expectation thanks to Lemma 1.3. The third term (second bracket) has zero expectation by properties of the χ^2_n distribution. □

The above result provides an unbiased estimate of the risk for nonlinear estimators. When σ^2 is known, it can for instance be used to choose the tuning parameters of Lasso or Elastic-Net estimators, as in this case explicit formulae for $\sum_{i=1}^n (\partial/\partial) f(y)$ are known.

1.3. **If the dimension is 3 or higher, the MLE can be improved upon uniformly for all unknown means.** We study the risk of estimators of the form

Theorem 1.2. *Let*

$$\hat{\mu} = f(y) = \Big(1 - \frac{\sigma^2 q}{\|y\|^2}\Big) y$$

for some deterministic $q > 0$. Then

$$\mathbb{E}_\mu[\|\hat{\mu} - \mu\|^2] = \mathbb{E}_\mu[\hat{r}_f] = n\sigma^2 - \mathbb{E}\Big[\frac{\sigma^4 q}{\|y\|^2}\big(2n - q - 4\big)\Big].$$

Proof. We first compute $\|y - f(y)\|^2$ and $\frac{\partial f_i}{\partial y_i}(y)$ separately: we have

$$\|y - f(y)\|^2 = \|\frac{\sigma^2 q}{\|y\|^2} y\|^2 = \frac{\sigma^4 q^2}{\|y\|^2}$$

as well as

$$\frac{\partial f_i}{\partial y_i}(y) = \Big(1 - \frac{\sigma^2 q}{\|y\|^2}\Big) + \frac{2\sigma^2 q y_i^2}{\|y\|^4}, \qquad 2\sigma^2 \sum_{i=1}^n \frac{\partial f_i}{\partial y_i}(y) = 2\sigma^2 n\Big(1 - \frac{\sigma^2 q}{\|y\|^2}\Big) + \frac{4\sigma^4 q}{\|y\|^2}.$$

Summing the three terms in $\hat{r}_f$, we obtain the claim. □

We now come back to the two values of q previously mentioned:

- If $q = n$, we obtain

$$\mathbb{E}_\mu[\|\hat{\mu} - \mu\|^2] = n\sigma^2 - \mathbb{E}\Big[\frac{\sigma^4 n}{\|y\|^2}\big(n - 4\big)\Big]$$

  so that $\hat{\mu}$ improves upon the MLE for all $n \geq 5$.
- For the variant $q = (n-2)$, which is the James-Stein estimator,

$$\mathbb{E}_\mu[\|\hat{\mu} - \mu\|^2] = n\sigma^2 - \mathbb{E}\Big[\frac{\sigma^4 (n-2)^2}{\|y\|^2}\Big]$$

  so that it improves upon the MLE for all $n \geq 3$.